C-DAC Bangalore

Project Pool – PGCP BDA Course

The Project Pool for the PGCP BDA Course mentioned in Table 1 has been strategically curated to provide a structured and outcome-driven approach to project-based learning. It comprises a total of 55 carefully designed project ideas, systematically categorized into three tiers based on complexity and execution depth. Tier 3 (18 projects) focuses on low-complexity projects aimed at building foundational skills, enabling students to gain confidence through achievable and well-scoped solutions. Tier 2 (20 projects) represents moderate complexity, balancing implementation effort with conceptual understanding and practical application. Tier 1 (15 projects) includes high-complexity problems that demand advanced system design, deeper technical integration, and strong analytical capabilities low-complexity projects aimed at building foundational skills, enabling students to gain confidence through achievable and well-scoped solutions.

This tiered architecture ensures inclusivity across varying skill levels while maintaining strong alignment with industry expectations. It is designed to progressively enhance students' problem-solving abilities, technical proficiency, and execution discipline, ensuring that every participant can deliver a meaningful and demonstrable project within the program timeline.

Table 1 Project pool for the PGCP BDA course

#	Tier	Project Title	Problem Statement	Proposed Solution	Key Features	Expected Learning Outcomes
1	3	Plant Disease Classifier from Leaf Images	Small-scale farmers cannot afford expert agronomists to diagnose crop diseases, leading to untreated outbreaks and yield loss.	Train a CNN on labelled leaf disease images (PlantVillage dataset) to classify disease type and severity from a photo upload.	Image upload and preprocessing pipeline; CNN model training with transfer learning (MobileNetV2); Top-3 disease prediction with confidence scores; Disease description and treatment suggestion lookup; Model performance dashboard (accuracy, confusion matrix); Export misclassified samples for retraining	How CNNs extract spatial features, transfer learning vs training from scratch, image classification pipeline end-to-end
2	3	Fake News Headline Binary Classifier	Readers cannot quickly distinguish credible headlines from fabricated ones, especially on social platforms where articles spread without context.	Fine-tune a pre-trained BERT model on a labelled fake/real news headline dataset to predict headline credibility with an explanation.	Text preprocessing and tokenisation pipeline; BERT fine-tuning on LIAR or FakeNewsNet dataset; Confidence score with real vs fake label; LIME-based explanation of top contributing tokens; Batch CSV scoring for editorial teams; Performance metrics dashboard	How transformer tokenisation works, fine-tuning vs training BERT, why attention explains predictions
3	3	Resume Skill Entity Extractor	HR teams manually scan hundreds of resumes to identify skill keywords, wasting hours per hiring cycle.	Build a Named Entity Recognition (NER) model using spaCy or Hugging Face to extract skills, experience, and education entities from resume text.	PDF/text resume parsing; Custom NER model trained on annotated resume corpus; Entity extraction: skills, tools, years of experience, education; Skill taxonomy mapping to a standard ontology (e.g., ESCO); Structured JSON output per resume; Batch processing for bulk resume folders	NER model training and annotation workflow, entity tagging schemes (BIO), custom NLP pipelines

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4	3	Customer Support Ticket Intent Classifier	Support teams manually read and route tickets to departments, causing delays when volume spikes.	Train a multi-class text classifier using fine-tuned DistilBERT to categorise incoming support tickets by intent and urgency.	Text cleaning and class balancing pipeline; DistilBERT fine-tuning for multi-label classification; Intent categories: billing, technical, complaint, general; Urgency scoring based on linguistic cues; Auto-routing output with department tag; Confidence threshold for human-review flagging	Multi-class vs multi-label classification, handling class imbalance in text data, deploying NLP classifiers
5	3	Voice Emotion Recognition from Audio Clips	Call centres have no automated way to detect when a caller is frustrated, leading to poor escalation timing.	Extract MFCC audio features from call recordings and train a CNN to classify emotional tone into angry, happy, sad, and neutral.	Audio loading and segmentation using librosa; MFCC, chroma, and spectral contrast feature extraction; CNN model trained on RAVDESS / TESS dataset; Real-time emotion label from 5-second audio window; Emotion trend chart across a call session; Confusion matrix and per-class F1 dashboard	Audio signal processing, MFCC intuition, applying CNNs to non-image data
6	3	Legal Document Extractive Summariser	Legal professionals spend hours reading lengthy contracts to find key clauses, obligations, and deadlines.	Apply extractive summarisation using sentence scoring (TF-IDF, TextRank) and an NLP pipeline to surface critical clauses from legal PDFs.	PDF text extraction and sentence segmentation; TextRank and TF-IDF hybrid sentence importance scoring; Clause-type labelling: payment, termination, liability, IP; Top-N sentence extraction by clause type; Side-by-side original vs summary view; Export summarised clauses to structured report	Extractive vs abstractive summarisation, graph-based NLP algorithms, working with unstructured legal text
7	3	Product Review Sentiment and Aspect Tagger	Product managers read thousands of reviews manually to understand which specific features customers love or hate.	Build an Aspect-Based Sentiment Analysis (ABSA) pipeline that identifies product aspects and assigns sentiment polarity to each.	Review scraping or CSV import; Aspect extraction using dependency parsing (spaCy); Sentiment polarity per aspect using VADER and BERT; Aspect frequency heatmap across reviews; Most-negative and most-positive review surfaces; Export to structured Excel for PM teams	Aspect-based NLP vs document-level sentiment, dependency parsing, combining rule-based and ML approaches
8	3	Hate Speech and Toxicity Detector for Social Posts	Community moderators on forums and social platforms cannot keep up with the volume of user-generated posts that may violate guidelines.	Fine-tune a DistilBERT or RoBERTa model on the Twitter Hate Speech dataset to flag toxic content across categories.	Social post text preprocessing; Fine-tuned transformer classifier for hate, offensive, and neutral classes; Multi-label output: toxicity, insult, threat, identity attack; Explanation of flagged content using attention weights; Moderation confidence score and threshold tuning; Bulk CSV evaluation for community managers	Ethical considerations in NLP classification, multi-label modelling, bias in hate speech datasets
9	3	Code Programming Language Identifier	Code snippet search engines and snippet-sharing tools lack automatic detection of programming language, requiring manual tagging.	Train a text classification model on GitHub code snippets to predict the programming language from syntax patterns alone.	Code tokenisation pipeline preserving syntax patterns; Multi-class classifier (15+ languages) using character-level CNN; Confidence score per prediction with top-3 alternatives; Syntax highlighting integration based on prediction; Edge case handling for mixed-language files; Model retraining pipeline for new languages	Character-level vs word-level NLP models, multi-class classification, tokenising code vs natural language

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10	3	Medical Symptom Text Triage Classifier	Health helplines receive free-text symptom descriptions that nurses must manually read to determine urgency level for routing.	Build a transformer-based text classifier trained on medical symptom datasets to assign urgency tiers: emergency, urgent, routine.	Medical text preprocessing with domain stopword handling; Fine-tuned BioBERT or ClinicalBERT for triage classification; Urgency tier output: emergency / urgent / routine; Highlighted symptom keywords driving the classification; Disclaimer and confidence threshold for safety; Evaluation on held-out real-world cases	Domain-specific pre-trained models, responsible AI in healthcare, calibration of high-stakes classifiers
11	3	News Article Reading Difficulty Scorer	Publishers and educators cannot quickly gauge whether a news article's language is accessible to a target audience age group.	Build an NLP pipeline combining classical readability formulas with a fine-tuned model to score article difficulty and suggest simplifications.	Article text ingestion via URL or paste; Flesch-Kincaid, Gunning Fog, and SMOG index computation; Sentence complexity scoring using dependency parse depth; Grade-level label mapping (elementary / intermediate / advanced); Top-5 hard sentences flagged for editing; Batch scoring for content review teams	Classical NLP metrics, combining rule-based and ML signals, linguistic feature engineering
12	3	Image-Based Document Type Classifier	Back-office teams in banks and insurance firms receive scanned documents in mixed batches and manually sort them before processing.	Train a CNN to classify scanned document images into types: invoice, ID card, bank statement, contract, and form.	Image preprocessing: deskew, binarise, normalise; CNN classifier (EfficientNetB0) trained on DocVQA / RVL-CDIP dataset; Top-3 document type prediction with confidence; Bounding box highlighting of classification-relevant regions; Batch folder processing pipeline; Misclassification logging for model feedback	Document image classification vs OCR, CNN transfer learning choices, production-level batch inference
13	3	Job Description Keyword Gap Analyser	Job seekers cannot tell which keywords in their resume are missing compared to target job descriptions, lowering ATS match rates.	Use NLP to extract skill entities from both a resume and a job description, then surface the gap as a prioritised recommendation list.	Dual text input: resume and job description; Skill extraction using NER and keyword matching; Gap analysis: skills present, missing, and partially matched; TF-IDF importance ranking of missing keywords; Recommended additions with context from job description; Export gap report to PDF	NLP-based information extraction, similarity metrics between documents, practical application of NER
14	3	Retail Sales Forecasting for Small Stores	Small retailers over-order or under-order stock because they rely on gut feel instead of historical demand patterns.	Train a time-series forecasting model (ARIMA + XGBoost hybrid) on weekly sales data to predict next 4-week demand per product category.	Historical sales ingestion and trend decomposition; Seasonality and holiday feature engineering; ARIMA baseline vs XGBoost comparison; Forecast confidence interval visualisation; SKU-level prediction output; RMSE and MAPE evaluation dashboard	Time-series fundamentals, feature engineering for temporal data, comparing statistical vs ML forecasting
15	3	Student Dropout Early Warning System	Colleges identify at-risk students only after they have already dropped out, leaving no window for intervention.	Build a logistic regression and random forest classifier on student engagement and academic data to score each student's dropout risk monthly.	Feature engineering from attendance, grades, and LMS activity; Logistic regression baseline with interpretable coefficients; Random forest with feature importance ranking; Monthly risk score per student; SHAP-based explanation of top risk drivers; Counsellor dashboard with flagged student list	Supervised classification, feature importance vs SHAP explanation, building ML for social good

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16	3	House Price Estimator with Explainability	Homebuyers and sellers have no transparent tool that explains why a property is valued at a certain price beyond a black-box number.	Build a gradient boosting regression model on public property datasets and surface SHAP-based explanations per prediction alongside comparable properties.	Property feature ingestion: location, size, amenities, age; Gradient boosting regressor (XGBoost / LightGBM) training; SHAP waterfall chart per property prediction; Comparable property lookup from training data; Prediction confidence interval display; Feature correlation heatmap for data exploration	Regression modelling, SHAP for tabular explainability, communicating ML outputs to non-technical users
17	3	Credit Card Default Risk Scorecard	Lenders approve or reject applicants using opaque scoring systems that applicants cannot understand or challenge.	Build an interpretable logistic regression scorecard on public credit data that assigns point-based risk scores with transparent reasoning per factor.	Credit feature preprocessing: income, utilisation, delinquency history; Logistic regression with Weight of Evidence encoding; Points-based scorecard generation per feature bin; Score-to-probability calibration chart; Population Stability Index for data drift monitoring; Applicant-level score breakdown report	Credit scoring methodology, WoE encoding, building interpretable ML vs black-box models
18	3	E-Commerce Product Return Predictor	Returns cost retailers 20–30% of revenue, yet no signal is acted on at the point of purchase to flag high-return-risk orders.	Train a binary classifier using order, product, and customer features to predict the probability of return at checkout, enabling targeted interventions.	Feature engineering: product category, price, review score, customer history; Class imbalance handling with SMOTE; XGBoost binary classifier with probability output; Threshold tuning for precision vs recall trade-off; Top-10 feature importance chart; Business impact calculation: estimated return reduction	Imbalanced classification, threshold selection, translating ML output into business value
19	2	Automated Meeting Transcript Summariser with Action Items	After every meeting, someone must manually write the summary and action items from a transcript, which takes 30–60 minutes and is inconsistent.	Build a GenAI pipeline using an LLM to ingest raw meeting transcripts and produce structured summaries, decisions, and timestamped action items.	Raw transcript ingestion (TXT, SRT, Whisper output); Speaker attribution and turn segmentation; LLM prompt chain for: summary, decisions, action items; Action item extraction with owner and deadline fields; Comparison of multiple summarisation prompts for quality; Exportable structured report in Markdown and JSON	Prompt engineering for structured extraction, LLM output reliability, multi-step NLP pipelines
20	2	RAG-Based Company Policy Q&A System	Employees ask the same policy questions repeatedly to HR because searching through long PDF handbooks is tedious and inefficient.	Build a Retrieval-Augmented Generation (RAG) system that indexes company policy PDFs and answers natural language queries using retrieved context.	PDF ingestion and chunking pipeline; Embedding generation using sentence-transformers; FAISS or ChromaDB vector store indexing; Retrieval of top-K relevant chunks per query; LLM answer generation grounded in retrieved context; Source citation with page reference for every answer	RAG architecture and retrieval mechanics, vector embeddings intuition, grounded vs hallucinated generation
21	2	AI-Powered Code Review Assistant	Junior developers receive slow feedback on code quality, readability, and potential bugs because senior engineers are a bottleneck.	Build a code review assistant using an LLM that analyses submitted code snippets and returns specific, line-level improvement suggestions.	Code snippet input via paste or GitHub PR URL; Language detection and syntax-aware chunking; LLM review for: bugs, readability, performance, security; Line-level comment generation with explanation; Severity tagging: critical / warning / suggestion; Diff-style output showing original and improved code	LLM prompting for structured technical output, code as text for NLP, building developer-facing AI tools

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22	2	Multilingual Customer Review Analyser	Global e-commerce platforms receive reviews in 10+ languages but only analyse English ones, missing critical signals from regional markets.	Build a pipeline using multilingual-BERT (mBERT) or XLM-RoBERTa to classify sentiment and extract topics across reviews in any language.	Language auto-detection per review; Multilingual sentiment classification using XLM-RoBERTa; Aspect-topic extraction across language pairs; Translation fallback using Helsinki-NLP models; Cross-language sentiment comparison dashboard; Regional sentiment heatmap by product category	Multilingual NLP, zero-shot vs fine-tuned cross-lingual transfer, handling noisy multilingual data
23	2	AI MCQ Generator from Study Material	Teachers spend significant time creating multiple-choice questions from textbook chapters, which reduces time for actual instruction.	Build a GenAI pipeline that takes educational text as input and generates distractors-included MCQs with answer keys and difficulty labels.	Text ingestion: PDF chapters, pasted notes, or URL; Key concept extraction using noun phrase and NER analysis; LLM-based question stem generation per concept; Distractor generation using semantic similarity search; Difficulty classification: easy / medium / hard; Export to GIFT format compatible with Moodle / Google Forms	Controlled generation with prompts, output evaluation strategies, building educational AI responsibly
24	2	Medical Report Plain-Language Simplifier	Patients receive medical reports filled with clinical jargon they cannot understand, increasing anxiety and reducing informed decision-making.	Build a GenAI pipeline using an LLM to transform medical reports into patient-friendly plain language while preserving all clinical facts.	Medical PDF/text ingestion and section parsing; LLM-based paraphrasing with medical jargon substitution; Fact consistency checker comparing original vs simplified version; Reading grade level score (Flesch-Kincaid) before and after; Highlighted medical terms with layperson definitions; Disclaimer injection for medical AI outputs	Controlled generation for sensitive domains, hallucination detection, LLM-as-evaluator pattern
25	2	Contract Clause Risk Analyser with NLP	Startups sign vendor and SaaS contracts without legal counsel, missing high-risk clauses around liability, data, and termination.	Build an NLP system that classifies contract clauses into risk categories and flags deviations from standard market language.	Contract PDF ingestion and clause segmentation; Clause-type classifier: liability, termination, IP, data, payment; Risk-level labelling: low / medium / high per clause; Deviation detection against a reference standard contract; LLM-based one-line risk explanation per flagged clause; Executive risk summary report export	Domain adaptation of NLP models, clause-level NLP vs document-level, building AI systems with legal constraints
26	2	Deep Learning Based Time-Series Anomaly Detector	Operations teams monitor thousands of sensor streams manually and miss early anomaly signals in equipment telemetry data.	Train an LSTM Autoencoder on normal operational time-series data to detect anomalies as reconstruction errors exceeding a threshold.	Time-series ingestion: CSV or streaming format; Sliding window normalisation and preprocessing; LSTM Autoencoder training on normal-class data only; Reconstruction error threshold selection with ROC analysis; Anomaly segment flagging with severity score; Real-time scoring simulation on held-out test stream	Autoencoders for anomaly detection, reconstruction error intuition, unsupervised deep learning for operations
27	2	AI-Powered Job Posting Bias Detector	Job descriptions written by recruiters unknowingly contain gendered or exclusionary language that narrows the applicant pool and affects diversity.	Build an NLP pipeline that scores job postings on bias dimensions (gender, age, ability) and suggests rewritten alternatives using an LLM.	Job description text input or CSV batch import; Bias lexicon matching for gendered and age-coded terms; ML classifier for overall bias tone score; Bias type tagging: masculine-coded, ageist, ableist; LLM-generated neutral rewrite for each biased phrase; Before/after bias score comparison dashboard	Bias in NLP systems, lexicon-based vs model-based approaches, responsible AI in HR applications

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28	2	Multi-Document Research Summariser	Analysts and researchers must read 10–15 papers or reports on a topic and manually synthesise insights, taking days instead of hours.	Build a GenAI pipeline that ingests multiple documents, extracts key claims, and synthesises a coherent thematic summary with source citations.	Multi-PDF ingestion with per-document parsing; Key claim extraction per document using LLM; Cross-document theme clustering using embedding similarity; Synthesised narrative summary with source attribution; Contradicting claims detection and flagging; Exportable structured research brief in Markdown	Multi-document NLP challenges, source attribution in GenAI, abstractive synthesis vs aggregation
29	2	AI Email Reply Draft Generator	Professionals spend 30–40% of their day writing repetitive reply emails that could be accelerated with context-aware drafts.	Build a GenAI tool that reads the incoming email thread and user's preferred tone to generate 3 draft reply options at different levels of formality.	Email thread ingestion and intent classification; Tone selector: formal / professional / casual; LLM-based draft generation (3 variants per intent); Key information auto-populated from email context; One-click edit and personalise before sending; Tone consistency scoring of the generated draft	Prompt engineering for stylistic variation, in-context learning, LLM controlled generation techniques
30	2	Visual Product Similarity Search Engine	E-commerce shoppers who see a product they like in a photo cannot find the same or similar product by text description alone.	Build a deep learning visual search engine using CNN embeddings and approximate nearest-neighbour search to return visually similar products.	Image upload from URL or device camera; Feature extraction using ResNet50 / EfficientNet embeddings; FAISS ANN indexing of product image catalogue; Top-5 visually similar product retrieval with similarity score; Combined visual + text filter refinement; Response time benchmarking under catalogue scale	Deep feature embeddings, ANN search structures, visual similarity vs keyword search, building real search systems
31	2	Personalised Content Recommendation using Embeddings	Content platforms show the same popular articles to all users instead of surfacing items aligned with individual reading history and interests.	Build a content recommendation system using user-content interaction embeddings trained with a two-tower neural network model.	User reading history and content metadata ingestion; Two-tower neural network for user and item embeddings; ANN-based retrieval of top-N personalised recommendations; Cold-start handling for new users via content metadata; A/B test simulation comparing embedding vs collaborative filtering; Explanation: top topics driving each recommendation	Neural recommendation systems, two-tower architecture, handling cold-start in real recommenders
32	2	Intelligent Threat Intel Extractor from Security Reports	Security analysts read lengthy threat intelligence reports and manually extract IOCs, TTPs, and threat actor details into spreadsheets.	Build an NLP pipeline using fine-tuned NER and an LLM extraction chain to automatically parse security reports and populate a structured threat database.	Security PDF / blog post ingestion; Custom NER for: IP addresses, CVEs, malware names, threat actors, TTPs; MITRE ATT&CK tactic and technique auto-tagging; LLM-based relationship extraction between entities; Structured JSON output per report; Threat summary dashboard with timeline view	Domain-specific NER fine-tuning, information extraction pipelines, applied NLP for cybersecurity
33	2	Video Lecture Transcript Q&A Bot	Students who watch long lecture recordings cannot quickly find answers to specific questions without rewatching the full video.	Build a RAG-based Q&A bot over Whisper-generated transcripts of video lectures, enabling students to ask questions and get timestamped answers.	Automatic transcription using OpenAI Whisper; Transcript chunking with timestamp preservation; Embedding and vector store indexing of transcript chunks; Natural language Q&A with answer grounded in transcript; Timestamp citation linking to exact video moment; Multi-lecture knowledge base with source identification	End-to-end RAG pipeline, audio-to-text preprocessing, building search over unstructured video content

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34	2	Social Media Trend Monitoring Agent	Brand managers and marketers monitor social media trends manually, missing early signals of emerging topics relevant to their product.	Build a lightweight agentic system that fetches social posts, clusters emerging topics using NLP, and generates a daily trend briefing automatically.	Social post ingestion via API (Reddit/Twitter) or RSS; BERTopic-based dynamic topic modelling and trend detection; Topic velocity tracking: growth rate over 24–48 hours; Sentiment polarity per trending topic; LLM-generated one-paragraph briefing per topic; Scheduled daily email digest with top trends	Dynamic topic modelling, combining NLP with automation, building scheduled agentic pipelines
35	2	Automated Data Quality Report Generator	Data engineers spend hours manually profiling new datasets for completeness, consistency, and distribution anomalies before analysis can begin.	Build an agentic system that ingests any tabular dataset, profiles it automatically, and uses an LLM to write a plain-English data quality report.	CSV/Excel ingestion with schema inference; Statistical profiling: nulls, distributions, outliers, cardinality; Schema drift detection vs a reference baseline; LLM narrative generation per data quality finding; Severity tagging: blocking / warning / advisory per issue; Exportable HTML data quality report with charts	Automated profiling pipelines, LLM for structured-to-narrative translation, agentic workflow design
36	**2**	Dynamic Customer Lifetime Value Segmentation Engine	Marketing teams use static RFM tiers that don't update as customer behaviour changes, causing misallocation of retention budgets.	Build a pipeline combining probabilistic CLV modelling (BG/NBD) with K-Means clustering to assign customers to dynamic value tiers that update weekly.	Transaction history ingestion and RFM feature computation; BG/NBD model for purchase frequency and churn probability; Predicted 12-month CLV per customer; K-Means clustering into 5 value tiers with weekly refresh; Tier migration tracking: customers moving up or down; Budget allocation recommendation per tier	Probabilistic ML models, combining unsupervised and supervised approaches, operationalising ML in marketing
37	2	Predictive Maintenance with Sensor Telemetry	Manufacturing plants run equipment to failure because reactive maintenance is cheaper only until unplanned downtime costs are accounted for.	Build a survival analysis and gradient boosting pipeline on sensor telemetry data to predict time-to-failure and schedule maintenance before breakdown.	Sensor data ingestion: vibration, temperature, pressure streams; Rolling window feature engineering (mean, std, trend slope); Cox Proportional Hazards model for time-to-failure estimation; XGBoost binary classifier for failure-within-N-days; Maintenance schedule recommendation output; Precision-recall evaluation with cost-weighted threshold selection	Survival analysis, feature engineering for time-series sensor data, cost-sensitive ML evaluation
38	2	Multi-Label Disease Prediction from Lab Reports	Patients presenting with multiple co-morbidities are often missed by single-disease screening models that don't account for co-occurring conditions.	Build a multi-label ML classifier on structured EHR and lab result data to predict the presence of multiple conditions simultaneously with calibrated probabilities.	EHR feature preprocessing and missing value imputation strategy; Multi-label binariser and label correlation analysis; Classifier chain vs binary relevance comparison; Calibrated probability output per disease label; AUROC per label with confidence interval; Model card with performance across demographic subgroups	Multi-label classification architectures, probability calibration, fairness evaluation in healthcare ML
39	2	Real Estate Investment Opportunity Scorer	Property investors evaluate deals manually using spreadsheets, missing data-driven signals about undervalued properties in large markets.	Build an ML pipeline that scores properties on investment attractiveness by combining price prediction residuals, neighbourhood trend features, and rental yield estimates.	Property listing ingestion from public dataset; Hedonic price model to identify under- and over-priced properties; Neighbourhood feature engineering: price growth trend, amenity density; Rental yield estimation using comparable rental data; Composite opportunity score with factor breakdown; Interactive map visualisation of opportunity scores	Feature engineering for real estate, residual-based anomaly scoring, multi-factor ML scoring systems

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40	2	Fraud Detection System with Concept Drift Handling	Financial fraud patterns shift as fraudsters adapt tactics, causing static ML models to degrade rapidly without retraining pipelines.	Build a fraud detection pipeline using gradient boosting with an automated drift detection and retraining trigger based on Population Stability Index and model performance monitoring.	Transaction feature engineering: velocity, geography, device, time-of-day; Imbalanced learning with cost-sensitive XGBoost; PSI-based input distribution drift monitoring; AUROC and KS-statistic model performance tracking; Automated retraining trigger when drift threshold breached; A/B champion-challenger model comparison framework	Concept drift detection, model monitoring in production, building ML systems that degrade gracefully
41	1	Multi-Agent Research and Synthesis Assistant	Researchers waste days gathering, reading, and synthesising information from multiple web sources because no tool both retrieves and reasons over content.	Build a multi-agent system using LangGraph or CrewAI with specialised agents: Researcher (web search), Summariser (extractive), Critic (fact-check), and Writer (final synthesis).	Orchestration layer using LangGraph agent graph with tool routing; Researcher agent with web search and PDF ingestion tools; Summariser agent with per-source key claim extraction; Critic agent comparing claim consistency across sources; Writer agent producing structured final report; Human-in-the-loop checkpoint before final report generation	Multi-agent system design, LangGraph orchestration patterns, building reliable AI pipelines with validation
42	1	Agentic SQL Data Analyst with Natural Language Interface	Business analysts without SQL skills cannot independently query databases, creating a bottleneck on data teams for every ad-hoc analysis request.	Build a conversational data analyst agent that translates natural language questions into SQL, executes queries, and returns chart-accompanied insights.	Natural language to SQL generation using LLM with schema context; SQL safety validation before execution (read-only guard); Query result parsing and chart type auto-selection; Follow-up question support with conversation memory; Error recovery loop: LLM self-corrects failed SQL; Exportable analysis session as reproducible notebook	Text-to-SQL architectures, agentic self-correction loops, tool use in LLM agents
43	1	Multimodal Invoice Processing and Accounting System	Finance teams manually extract line items, totals, and vendor details from hundreds of PDF invoices per month, an error-prone and time-consuming process.	Build a multimodal GenAI pipeline that uses a vision LLM to extract structured data from invoice images and maps them to an accounting schema.	Invoice PDF and image ingestion with layout detection; Vision LLM extraction: vendor, items, amounts, dates, tax; Confidence scoring per extracted field; Schema validation and anomaly flagging (e.g., total mismatch); Duplicate invoice detection using fuzzy matching; Structured JSON to accounting export format (CSV/Xero-compatible)	Multimodal LLM prompting, structured extraction from unstructured visual documents, validation pipelines
44	1	Autonomous Code Debugging and Repair Agent	Developers spend significant time debugging runtime errors and logic bugs, especially in unfamiliar codebases where tracing is complex.	Build an agentic system that accepts buggy code and an error trace, autonomously proposes fixes, tests them in a sandboxed executor, and iterates until tests pass.	Code ingestion with error trace parsing; LLM-based root cause hypothesis generation; Patch proposal with diff generation; Sandboxed code execution engine for automated testing; Self-correction loop: agent re-evaluates and re-patches on test failure; Explanation of the bug and fix for developer learning	Agentic self-correction loops, tool use with code execution, building reliable code AI with safety guards
45	1	Personalised AI Learning Path Generator	Online learners face choice overload when navigating course platforms and lack a personalised roadmap aligned to their goals and current skill gaps.	Build an agentic system that conducts a skill assessment dialogue, maps the learner's profile, and generates a dynamic, week-by-week learning path with curated resources.	Conversational skill assessment chatbot with adaptive questions; Skill gap extraction from assessment dialogue; Learning path generation grounded in a curated resource database; Weekly milestone planning with checkpoint tasks; RAG over course descriptions to recommend specific resources; Learning path adaptation based on progress feedback	Dialogue management, RAG for recommendation, combining assessment and generation in one system

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46	1	Real-Time News Intelligence Briefing Agent	Executives and analysts need a concise daily intelligence briefing but current tools either surface too much noise or lack synthesis across sources.	Build an agentic pipeline that fetches articles from 10+ sources, clusters them by topic, scores relevance to the user's domain, and generates a synthesised briefing.	Multi-source RSS/API news ingestion on schedule; BERTopic or LLM-based cross-source topic clustering; Relevance scoring against user-defined domain profile; LLM synthesis of top-5 briefing items with contradiction flags; Importance ranking using entity prominence and source credibility; Daily email delivery with clickable source citations	Agentic scheduling, multi-source synthesis, relevance ranking for GenAI systems
47	1	Agentic Customer Journey Automation System	E-commerce teams manually design and trigger customer re-engagement workflows, making it impossible to personalise at scale across different user behaviour patterns.	Build a multi-step agentic system that monitors customer event signals, reasons about customer intent, and autonomously triggers personalised outreach with LLM-generated content.	Customer event stream ingestion: browsing, cart, purchase; Intent classification agent using event sequence modelling; Decision agent: which intervention to trigger per customer segment; LLM-generated personalised email/SMS content per customer; Campaign performance feedback loop updating decision rules; Human approval gate for high-value interventions	Event-driven agentic design, combining ML classification with LLM generation, building closed-loop AI systems
48	1	Generative AI for Financial Earnings Report Analyst	Equity analysts spend hours reading 80-page quarterly earnings reports to extract key metrics, risks, and guidance before writing their investment summaries.	Build a GenAI pipeline that ingests SEC 10-Q/10-K filings and generates structured analyst summaries with key metric extraction, management tone analysis, and guidance parsing.	SEC EDGAR filing ingestion and section parsing (MD&A, Risk, Financials); LLM extraction of KPIs, guidance ranges, and risk factors; Management tone and language shift analysis vs prior quarter; Comparison table auto-generation: current vs prior period; Bull/bear signal surfacing with supporting evidence; Investor-grade PDF summary report generation	Long-document GenAI strategies, structured extraction from financial text, building domain-specific LLM applications
49	1	Multi-Modal Medical Image Report Generator	Radiologists face high workloads generating written reports for chest X-rays and scans, creating bottlenecks in patient care workflows.	Build a multimodal AI pipeline combining a vision encoder (CheXNet) with an LLM text decoder to generate structured preliminary radiology report drafts.	Chest X-ray image ingestion and preprocessing; CheXNet-based pathology classification: pneumonia, effusion, cardiomegaly; Vision feature extraction and LLM-compatible encoding; LLM decoder generating structured FINDINGS and IMPRESSION sections; Confidence score per finding with attention map overlay; Mandatory disclaimer and human-review flagging before output display	Vision-language model architecture, responsible AI in medical applications, structured text generation from image features
50	1	Adaptive Document Intelligence Agent with Memory	Knowledge workers dealing with large document repositories have no tool that can reason across multiple documents over time, remembering prior questions and building understanding.	Build an agentic document intelligence system with persistent memory that allows users to have ongoing, context-aware conversations across a growing document corpus.	Persistent user memory store per session and across sessions; Multi-document vector index with source metadata; Agentic retrieval planner: deciding how many and which docs to search; Cross-session context recall: agent remembers prior queries and answers; Contradiction detection between documents with source attribution; Memory reset and privacy controls for sensitive document sets	Agent memory architectures, persistent RAG systems, multi-session context management in production AI
51	1	Causal Uplift Modelling for Marketing Interventions	Marketing teams measure campaign ROI using pre/post comparisons that confuse correlation with causation, leading to budget wasted on customers who would have converted anyway.	Build a causal ML pipeline using uplift modelling (T-Learner, S-Learner, X-Learner) to estimate the incremental impact of a marketing treatment per customer segment.	Experimental data ingestion with treatment and control flags; T-Learner, S-Learner, and X-Learner uplift model training; Qini curve and uplift AUC evaluation; Persuadable vs sleeping-dog vs lost-cause customer segmentation; Budget optimisation: which customers to target for maximum incremental lift; Sensitivity analysis on treatment effect estimates	Causal inference in ML, uplift modelling frameworks, why average treatment effects mislead targeting decisions

#	Tier	Project Title	Problem Statement	Proposed Solution	Key Features	Expected Learning Outcomes
52	1	AutoML Pipeline Builder with Meta-Learning	Data science teams spend days manually selecting algorithms, tuning hyperparameters, and comparing pipelines for each new dataset, making fast experimentation impossible.	Build an AutoML system that uses meta-learning over dataset characteristics to warm-start algorithm selection and applies Bayesian optimisation for hyperparameter search.	Dataset meta-feature extraction: statistical, information-theoretic, landmarking features; Meta-model trained on OpenML benchmark performance records; Warm-started algorithm ranking per new dataset; Bayesian optimisation with Optuna for hyperparameter tuning; Ensemble construction from top-K pipelines; Performance comparison report across all evaluated configurations	Meta-learning principles, Bayesian optimisation vs grid search, building reusable ML infrastructure
53	1	Graph-Based Fraud Network Detection System	Financial fraud often involves coordinated rings of accounts working together, yet transaction-level ML models miss the relational structure between entities.	Build a Graph Neural Network (GNN) pipeline that models customer-transaction-device relationships as a graph and detects fraud rings using node classification and community detection.	Graph construction from transactions: account, device, IP, merchant nodes; GraphSAGE or GAT node embedding training; Node-level fraud classification using graph features + tabular features; Community detection to surface coordinated fraud rings; Subgraph visualisation of flagged fraud clusters; Comparison vs tabular-only XGBoost baseline	Graph Neural Networks, combining relational and tabular signals, fraud ring detection vs individual fraud
54	1	Reinforcement Learning Agent for Dynamic Pricing	E-commerce platforms set prices using static rules that cannot adapt in real-time to inventory levels, competitor moves, and demand elasticity simultaneously.	Build a Reinforcement Learning agent using Q-Learning or PPO that learns a dynamic pricing policy by interacting with a simulated market environment built from historical data.	Historical sales and price data for environment simulation; Reward function design: revenue vs volume vs margin trade-off; Q-Learning baseline for discrete price actions; PPO agent for continuous pricing with stable training; Policy evaluation across demand shock and competitor scenarios; Safe exploration constraints to prevent extreme price actions	Reinforcement learning fundamentals, reward shaping, building RL environments from real business data
55	1	Federated Learning System for Privacy-Preserving Prediction	Healthcare and finance institutions cannot pool data to train better models due to privacy regulations, but isolated models trained on small local datasets underperform.	Build a federated learning simulation where multiple client nodes train local models on partitioned data and a central server aggregates updates using FedAvg without raw data ever leaving each node.	Data partitioning into IID and non-IID client distributions; Local model training with differential privacy noise injection; FedAvg aggregation server implementation; Global model convergence tracking across communication rounds; Performance comparison: federated vs centralised vs local-only models; Privacy-utility trade-off analysis across epsilon values	Federated learning architecture, FedAvg algorithm, differential privacy fundamentals, privacy-utility trade-offs in ML

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Mr. Anshul Tailor
Project Engineer, C-DAC Bangalore